



MRK HOTELS

Developer Documentation

Multi-tenant hotel management system

MRK Hotels — Developer Documentation

Version 1.4 — Architecture, Setup, Backend, Frontend and API Reference

1. Overview

MRK Hotels is a **multi-tenant hotel management SaaS** built as two applications:

Application	Stack	Location
API backend	Laravel 13 (PHP 8.3+), Sanctum, spatie/laravel-permission	mrk-hotels-api
Web frontend	Vue 3, Vite, Pinia, Vue Router, vue-i18n, Axios	mrk-hotels-frontend

The system has three faces:

1. **Public booking portal** — guests browse hotels, check availability and book rooms online, optionally paying via Selcom, mobile money (ClickPesa) or bank transfer.
2. **Hotel panel (/app)** — every hotel runs its own operation: reservations, rooms, guests, payments, housekeeping, food & beverage, laundry, fun & games, inventory, procurement, staff, messaging and reports.
3. **Superadmin panel (/superadmin)** — manages all tenants (hotels), approves registrations and toggles per-hotel payment methods.

Multi-tenancy is implemented at the application layer: models carry `tenant_id` and a global scope (`BelongsToTenant`), and a `TenantContext` support class scopes queries for the active hotel.

2. Requirements

Backend

- PHP ^8.3 (tested on 8.4)
- Composer
- MySQL / MariaDB (tests use in-memory SQLite)

Frontend

- Node ^22.18 / >= 24.12
- npm

Key dependencies

- Backend: `laravel/framework` ^13.0, `laravel/sanctum` ^4.0, `spatie/laravel-permission` ^7.3, `laravel/reverb` (**WebSocket server**)

- Frontend: vue, vue-router, pinia (+ persistedstate), vue-i18n, axios, font-awesome, country-state-city, libphonenumber-js, **laravel-echo** + **pusher-js** (**real-time client**); Vitest + Playwright for testing

3. Local Setup

3.1 Backend (mrk-hotels-api)

```
composer install
cp .env.example .env          # set DB_*, SANCTUM_STATEFUL_DOMAINS, CLICKPESA_*, REVERB_*
keys
php artisan key:generate
php artisan migrate --seed    # TenantSeeder creates the demo hotel + users + rooms
php artisan serve             # http://localhost:8000
php artisan reverb:start      # realtime WebSocket server (http://localhost:8080)
```

3.2 Frontend (mrk-hotels-frontend)

```
npm install
cp .env.example .env          # VITE_API_URL=http://localhost:8000/api
npm run dev                   # http://localhost:5173
```

3.3 Demo data (TenantSeeder)

- Hotel **MRK Grand Hotel** (Dodoma, Tanzania, **subdomain=mrk-grand** , status **active**) with demo tax IDs (**tin=123-456-789** , **vrn=40-012345-6**) and demo payment accounts (M-Pesa + NMB) so invoices render the full formal layout.
- 10 staff users (hotel_admin, manager, accountant, receptionist, procurement_officer, housekeeping, kitchen, waiter, bartender, staff) — all password **password** .
- 7 rooms (single ×2, double, deluxe ×2, suite, presidential).

A superadmin user (**superadmin@mrkhotels.test**) is created elsewhere (permissions seeder) with full access.

4. Backend Architecture

4.1 Authentication (Sanctum)

App\Http\Controllers\Api\V1\Auth\AuthController::login :

- Finds the user by email, verifies **password_hash** with **Hash::check** .
- Rejects disabled accounts (403) and inactive tenants (403).

- Handles **password rotation**: if `passwordExpired()`, the password is reset to the default (`defaultPassword()` = full name in caps) and returned with `password_rotated: true` and a 401, prompting the client to re-sign-in.
- Issues a token:
`createToken('api-token', $user->isSuperadmin() ? ['*'] : [$user->user_role])` — **abilities are the user's role name**.
- Returns `{ token, user }`.

`AuthController::loginPin` (`POST /api/v1/auth/login-pin`) — iPOS-style quick sign-in for shared staff terminals:

- Accepts `identifier` + `pin` (`digits:4`); the identifier matches the user's `email` **or** `registration_number`.
- Unknown identifiers, unset PINs and wrong PINs all get one generic 401 (`Invalid credentials.`) so the endpoint cannot be used to enumerate staff accounts.
- Applies the same gates as password login (active account, active/pending tenant), updates `last_login`, issues the identical token shape and audit-logs the login.
- PINs live bcrypt-hashed in `users.login_pin` (nullable — `null` disables PIN login for the account) and are hidden from serialization.
- PINs are assigned by senior staff via `UserController@setPin` (`POST /api/v1/users/{id}/set-pin`, level-80 group): validates `pin` + `pin_confirmation` (`digits:4`, `confirmed`), enforces `assertCanManage` on the target's role, rejects self-service with 422 (a profile self-service flow is planned), and audit-logs `pin_set`.

Protected routes use `auth:sanctum`. Token abilities are currently informational; authorization is enforced by the `level:X` and role-permission middleware below.

4.2 Authorization middleware

Middleware	Purpose
<code>CheckRoleLevel</code> (<code>level:X</code>)	Rejects users whose <code>roleLevel()</code> < X with 403 (superadmin bypasses). E.g. <code>level:80</code> = admin/manager only, <code>level:60</code> = receptionist+.
<code>RequireSuperadmin</code>	Superadmin-only routes.
<code>RequireOwner</code> (<code>owner</code>)	Multi-hotel owner routes (<code>/owner/*</code>): consolidated dashboard and per-hotel drill-down across the owner's hotels (<code>tenants.owner_id</code>). Owner accounts live on the platform tenant; hotel queries bypass global scopes with explicit <code>tenant_id</code> filters.
<code>SetTenantContext</code> (<code>tenant</code>)	Resolves the active tenant for the request: superadmin supplies <code>X-Tenant-ID</code> ; staff use their own <code>tenant_id</code> . Sets <code>TenantContext</code> for the request lifecycle.
<code>EnsureTenantActive</code>	Blocks requests for hotels whose tenant status is not active/pending.

Roles: `superadmin` (100), `owner` (95), `hotel_admin` (90), `manager` (80), ... Owners are created by the superadmin (`POST /api/v1/owners`, default password = full name in caps) and assigned to

hotels via `PUT /api/v1/tenants/{id}` (`owner_id`). Demo owner: `owner@mrkhotels.test` / `password` (owns MRK Grand, Dar es Salaam and Zanzibar).

4.3 Multi-tenancy

- The `BelongsToTenant` model concern adds a `tenant_id` foreign key and a **global scope** that filters queries to the current tenant (when `TenantContext::id()` is set).
- `App\Support\TenantContext` is a static holder set by the `tenant` middleware and cleared in a `finally` block.
- Public portal endpoints operate **without** a tenant context and pass an explicit `hotel_id` / `tenant_id`.

4.4 Routing (`routes/api.php`)

All routes are under the `v1` prefix.

Public group (`/v1/public`) — no auth:

Method	URI	Controller action
GET	<code>public/hotels</code>	<code>PublicController@hotels</code> — directory search (<code>search</code> , <code>country</code> , <code>city</code> , <code>guests</code> , <code>check_in/out</code>)
GET	<code>public/hotels/{id}</code>	<code>PublicController@hotelShow</code>
GET	<code>public/availability</code>	<code>PublicController@availability</code> — returns available rooms + <code>payment_methods</code>
GET	<code>public/options</code>	<code>PublicController@options</code>
POST	<code>public/booking-requisitions</code>	<code>PublicController@storeBookingRequisition</code>
POST	<code>public/reservations</code>	<code>PublicController@storeReservation</code> — creates pending reservations under one <code>booking_reference</code>
POST	<code>public/payments/initiate</code>	<code>PublicController@initiatePayment</code> — online payment entry point
GET	<code>public/booking-requisitions/status</code>	<code>PublicController@bookingStatus</code>

Public auth: `POST auth/login` , `POST auth/login-pin` , `POST auth/register` (all throttled by the shared `auth` limiter).

Payment webhook: `POST payments/clickpesa/callback` — no auth, used by the ClickPesa sandbox.

Authenticated group — `auth:sanctum` + `tenant` :

Module	Route prefix	Level
Auth	<code>auth/logout</code> , <code>auth/me</code> , <code>auth/change-password</code>	all

Module	Route prefix	Level
Reports	reports/dashboard , overview , occupancy , revenue , room-status , audit-logs	20 / 80 / 60 / 70 / 60 / 70
Messaging	messages/users , messages/unread-count , messages/conversations , messages/conversations/{id} , .../messages , .../send , .../read , .../delete , .../react , messages/groups , messages/groups/{id} , .../messages , .../send , .../read , .../members , .../delete , .../react , messages/statuses , messages/calls , messages/calls/{id}/actions plus the 24 feature endpoints: replies/priority/polls, pin/star, templates, scheduled messages, forwarding, announcements, search, CSV export, translate, escalations, retention policies, notification preferences/DND, shift handovers, room-linked chats, task groups, staff presence (location/nearby), guest SMS bridge, meetings, SOS alerts	20+
Users (staff)	users CRUD + activate/invite/reset-password/set-pin/attachments	80
Rooms	rooms CRUD + status	60
Guests	guests CRUD	60
Reservations	reservations CRUD + options + check-in/check-out/no-show/cancel + reservations/{id}/payment	60
Payments	payments + store	60
Booking requisitions	booking-requisitions	60
Housekeeping	housekeeping/tasks + room status	40
F&B	menu-items , orders	30+
Laundry	laundry-orders	40
Fun & games	fun-game-orders	80
Inventory	inventory + movements	50
Suppliers	suppliers CRUD	50
Procurement	requisitions , purchase-orders , goods-received-notes	50
Tenants	tenants CRUD + payment-methods	superadmin
Superadmin reports	reports/superadmin/dashboard	superadmin

(246 Route:: definitions total — 97 GET, 113 POST, 15 PUT, 20 DELETE, 1 PATCH.)

4.5 Models and relationships

Model	Key fields / notes
Core\Tenant	tenant_id (UUID), hotel_name , subdomain , status (pending/active/suspended), subscription_* , tax IDs (tin , vrn), invoice branding (signature_path , stamp_path on the public disk), payment_methods / payment_accounts (JSON casts) , enabledPaymentMethods() / publicPaymentMethods()
Auth\User	UUID user_id , tenant_id , user_role , password_hash , is_active , password_changed_at ; HasApiTokens, HasRoles
Hotel\Room	room_number , room_type , floor , price_per_night , max_occupancy , status
Hotel\Reservation	tenant_id , guest_id , room_id , status , booking_reference , dates, guest snapshot (denormalized first_name , guest_phone ,...)
Hotel\Guest	guest register, contact + ID info
Finance\Invoice	reservation_id , invoice_number , room_charges , service_charges , total_amount , paid_amount , balance , line_items / payments (JSON snapshots), status (issued/paid)
Finance\Payment	reservation_id , amount , payment_method , payment_provider , transaction_reference , status , booking_reference
System\BookingRequisition	online/phone booking requests (no room)
Housekeeping\HousekeepingTask , Laundry\LaundryOrder , Fun\FunGameOrder , FoodBeverage\Order / OrderItem / MenuItem , Inventory\InventoryItem / StockMovement / Supplier , Procurement* (requisition/PO/GRN + items)	domain records per module

Model	Key fields / notes
Messaging\Conversation / ConversationMessage , Messaging\GroupConversation / GroupConversationMember / GroupConversationMessage	staff messaging; conversation_messages carries delivered_at , type (text/audio/image/file), media_path , media_mime , view_once , is_media_seen , deleted_for , deleted_at , priority (normal/urgent), reply_to_id (self-join for replies), forwarded_from_type / forwarded_from_id (forwarding); group members track last_read_at for read receipts; message reactions stored as JSON (reactions column)
Messaging\PinnedMessage / StarredMessage	one row per pinned/starred message (message_type = conversation/group, message_id , user_id)
Messaging\Poll / PollOption / PollVote	message_type + message_id polymorphic polls; one vote per user per option
Messaging\MessageTemplate	reusable message snippets (name , body , category , is_global)
Messaging\ScheduledMessage	future-dated send (recipient_type = conversation/group, recipient_id , send_at , recurrence) — dispatched by the messaging:send-scheduled command
Messaging\Announcement / AnnouncementAcknowledgement	tenant-wide broadcast with per-user read acknowledgements
Messaging\MessageEscalation	raise a message to management (message_type , message_id , escalated_by , handled_by / handled_at)
Messaging\RetentionPolicy	scope (global/conversation/group), days , enabled — drives the messaging:purge-expired command
Messaging\NotificationPreference	per-chat mute + DND window (dnd_from / dnd_to), muted_until / muted , push_enabled , scope / target_id polymorphic
Messaging\ShiftHandover	shift change notes (shift_from / shift_to timestamps, status , ack_user_id / ack_at)
Messaging\ConversationRoom	links a chat (conversation_id or group_id) to a Hotel\Room
Messaging\StaffLocation	staff presence (zone , floor) with updated_at recency for the nearby lookup
Messaging\GuestMessage	outbound SMS bridge to guests (phone , provider , provider_status , template_id , direction)

Model	Key fields / notes
Messaging\Meeting / MeetingInvitee	scheduled meetings (<code>start_at</code> , <code>duration_minutes</code> , <code>conference_type</code> , <code>status</code>) with per-invitee responses
Messaging\SosAlert	emergency alert (<code>message</code> , <code>status</code> , <code>ack_count</code> , <code>ack_user_ids</code> , <code>resolved_by</code> / <code>resolved_at</code>); a <code>staff.location.updated</code> broadcast powers the nearby panel
Social\Status , Social\StatusView , Social\StatusReaction	24-hour status posts (<code>type</code> : text/photo/video), media under <code>statuses/{id}</code> ; <code>status_views</code> dedupe views per user; <code>status_reactions</code> holds emoji likes on statuses
Communication\Call , Communication\CallEvent	WebRTC calls: <code>type</code> (audio/video), <code>initiator_id</code> , <code>recipient_id</code> , <code>channel</code> (1:1 room name), <code>status</code> (ringing/active/missed/ended/rejected/declined), <code>ring_started_at</code> / <code>accepted_at</code> / <code>ended_at</code> ; <code>call_events</code> logs ring/accept/end/mute events with timestamps

4.6 Services

Service	Responsibility
Hotel\ReservationService	Room availability check, room totals (with optional ClickPesa service fee), reservation lifecycle (create, check-in/out, no-show, cancel).
Finance\PaymentService	Record payments, <code>initiatePublicPayment</code> (selcom → completed + confirms booking; bank → awaiting_confirmation; mobile money → ClickPesa prompt), confirm sibling reservations by <code>booking_reference</code> .
Finance\InvoiceService	<code>generateForReservation</code> builds/refreshes the folio invoice (room + non-cancelled F&B/laundry charges minus completed payments, frozen into JSON snapshots; regenerating keeps the invoice number). <code>pdf()</code> renders <code>pdf.invoice</code> via dompdf with the MRK logo, tenant branding (signature/stamp), TIN/VRN, VAT-inclusive 18% breakdown, payment terms and "how to pay" lines — images embedded as data URIs.
Inventory\InventoryService	Stock adjustments and movements.
System\AuditService	Audit trail for logins and key actions.

Service	Responsibility
Messaging\ConversationController / GroupConversationController	Delivery/read ticks: mark messages <code>delivered_at</code> on poll (<code>index/show/messages</code>), mark <code>read_at</code> on <code>markRead</code> ; media upload (public disk, ≤ 10 MB, stored under <code>messages/conversations/{id}</code> / <code>messages/groups/{id}</code> , type inferred from MIME); group <code>read_by_count</code> computed in <code>GroupMessageResource</code> ; delete (<code>delete_for</code> = <code>me/everyone</code>), reactions (emoji toggle), view-once media (media URL served once, then <code>is_media_seen</code>); replies (<code>reply_to_id</code> — <code>ReplySent</code> broadcasts the full reply resource), priority (<code>normal</code> / <code>urgent</code>), polls (poll payload: <code>question</code> , <code>multiple</code> , <code>options[]</code>)

Service	Responsibility
Messaging feature controllers	PinStarController (pin/unpin/star/unstar + pinned / starred lists, broadcasts MessagePinned / MessageUnpinned / MessageStarred / MessageUnstarred), PollController (vote once per user; broadcasts PollCreated / PollVoted with live tallies), MessageTemplateController (CRUD, optional category), ScheduledMessageController (index/store/cancel), AnnouncementController (post + per-user acknowledge; broadcasts AnnouncementPosted / AnnouncementAcknowledged on the tenant channel), SearchController (keyword search across chats the caller can access, ConversationMessage + GroupConversationMessage), ExportController (CSV export of a conversation/group history), TranslateController (offline EN↔SW glossary TranslateService , no external API), EscalationController (raise + resolve; broadcasts MessageEscalated ; auto-escalation of urgent unread messages via the messaging:escalate command (default unread threshold 10 minutes)), RetentionPolicyController (CRUD + messaging:purge-expired command soft-deletes expired messages), NotificationPreferenceController (per-chat mute + DND + push), ShiftHandoverController (post + acknowledge; broadcasts ShiftHandoverPosted / ShiftHandoverAcknowledged), RoomLinkController (searchRooms / link / unlink / index — links a chat to a hotel room), TaskGroupController (store + convert a room-linked message into a HousekeepingTask ; broadcasts TaskConverted), StaffLocationController (update zone/floor + nearby query; broadcasts StaffLocationUpdated), GuestMessageController (outbound SMS bridge, tenant-scoped), MeetingController (searchUsers , schedule + invitees respond; broadcasts MeetingInvited / MeetingResponseChanged on the user channel), SosController (initiate/acknowledge/resolve; broadcasts SosAlertInitiated / SosAlertResolved), ForwardController (copy a message into another conversation/group with forwarded_from_* provenance; validates access on source and target).
Console commands	messaging:send-scheduled (queued sends due at send_at), messaging:purge-expired (retention + scheduled cancelled_at cleanup), messaging:escalate (auto-escalate urgent messages unread past a threshold).
Social\StatusController	Post (text/photo/video), list/feed (own + others, auto-expires after 24 h), view/like. Broadcasts StatusPosted to the tenant channel so conversations show live status rings.

Service	Responsibility
Communication\CallController	WebRTC call lifecycle: start (creates Call + rings via CallStarted broadcast), accept , reject , decline (busy), end , miss (timeout), plus a ping event for live ring/ringing state.
Realtime events	MessageSent (body/media/type/view_once/priority/reply), MessageRead , MessageDeleted (includes channel_id), MessageReacted , ReplySent , MessagePinned / MessageUnpinned , MessageStarred / MessageUnstarred , PollCreated , PollVoted , TaskConverted , AnnouncementPosted , AnnouncementAcknowledged , SosAlertInitiated , SosAlertResolved , ShiftHandoverPosted , ShiftHandoverAcknowledged , StaffLocationUpdated , GuestMessagePosted , MessageEscalated , MeetingInvited , MeetingResponseChanged , StatusPosted , CallStarted , CallEvent , MemberJoined / MemberLeft — all on private channels (private-users.{id} , private-tenants.{tenant_id} , private-conversation.{id} , private-group.{id}) via Reverb.

4.7 Support classes

- `PaymentOptions` — methods (`cash` , `mobile_money` , `bank` , `selcom` , `card`), `ONLINE_METHODS` (`selcom` , `mobile_money` , `bank`), providers (mobile money: `airtel_money` , `mixx_by_yas` , `halopesa` , `mpesa` ; bank: `crdb` , `nmb` , `nbc` , `other`), statuses (`pending` , `awaiting_confirmation` , `completed` , `failed` , `refunded`), and helpers (`defaultMethods()` excludes `selcom`, `requiresProvider` , `requiresConfirmation` , `initialStatus` , `label()` for document rendering).
- `BookingOptions` — booking types (single, couple, family, group), room types, ID types, booking sources, guest defaults, `resolveStay()` for nights/departure.
- `NumberGenerator` — per-tenant sequential numbers: `prNumber` , `poNumber` , `grnNumber` , `orderNumber` , `requisitionNumber` , **`bookingReference (BK-...)`** , `staffNumber` , `laundryNumber` , `funGameNumber` , `transactionReference` , `clickPesaReference` , `inviteToken` , `tempPassword` .
- `TenantContext` — static tenant id holder.
- Module options: `HousekeepingOptions` , `LaundryOptions` , `FunGameOptions` , `OrderOptions` , `StaffOptions` .

4.8 Public booking & payment flow

1. `GET public/availability?hotel_id&check_in&check_out&booking_type` returns available rooms **and** the hotel's enabled `payment_methods` .
2. `POST public/reservations` validates the guest + booking (including `booking_date` after today and before check-in). With `room_selections` , one **pending** `Reservation` is created per room under a shared `booking_reference (BK-...)` ; the response carries `booking_reference` and `total_amount` (room rate × nights + ClickPesa service fee when online). Without rooms it creates a booking requisition.

3. `POST public/payments/initiate` with `booking_reference` , `amount` , `payment_method` , `payment_provider` , `phone` :
4. **selcom** → payment `completed` , all same-reference pending reservations confirmed.
5. **bank** → payment `awaiting_confirmation` ; hotel confirms later.
6. **mobile_money** → ClickPesa prompt via `phone` ; the `payments/clickpesa/callback` webhook completes the payment and confirms the booking.

4.9 Migrations & seeders

- ~67 migrations cover tenants, users (incl. `add_login_pin_to_users_table` — the nullable, hashed 4-digit `users.login_pin` powering PIN sign-in), permissions, rooms, guests, reservations, payments, booking requisitions, housekeeping, F&B, laundry, fun & games, inventory, procurement, staff invitations/attachments, audit logs, **messages (reactions/view-once/delete fields), statuses (statuses / status_views / status_reactions), calls (calls / call_events)** plus the messaging feature set: `pin_star_templates_scheduled_tables` (pinned/starred/templates/scheduled), `polls_announcements_tables` (polls/poll_options/poll_votes/announcements + acknowledgements), `preferences_retention_escalation_handover_tables` (notification preferences, retention policies, escalations, handovers), `task_groups_staff_locations_guest_messages` (conversation_rooms, task_groups, staff_locations, guest_messages) and `meetings_sos_tables` (meetings, meeting_invitees, sos_alerts + `ack_user_ids`).
- `TenantSeeder` seeds the demo tenant; `RolePermissionSeeder` sets up roles/permissions.

4.10 Testing

- `phpunit.xml` uses in-memory SQLite (`:memory:`) for tests.
- Base `Tests\TestCase` with `RefreshDatabase` .
- Feature tests include `tests/Feature/PublicBookingPaymentFlowTest.php` (7 tests: availability methods, selcom default-off, pending booking creation, selcom confirms, bank awaiting-confirmation, selcom rejected when disabled, multi-room shared reference + batch confirm), `tests/Feature/ConversationTest.php` (direct messaging: scoping, resumes, unread counts, polling, delivery/read ticks, audio upload), `tests/Feature/GroupConversationTest.php` (11 tests: hotel/global groups, member guards, read receipts, media, add/remove/leave, unread badge feed), `tests/Feature/MessagingFeaturesTest.php` (**20 tests** covering the 24-feature set: replies + priority + polls, poll voting, pin/star, templates, scheduled messages, announcements + acknowledge, escalation + resolve, command auto-escalation, search, CSV export, translate, task-group conversion, meetings + responses, SOS flow, staff location/nearby, guest SMS, retention purge, **forwarding (incl. cross-chat access block), room search, meeting-invitee search**), `tests/Feature/OverviewPaginationTest.php` (3 tests: staff/in-house/housekeeping section filtering + pagination), `tests/Feature/InvoiceTest.php` (folio totals, regeneration keeps number and settles, PDF download, front-desk level, tenant scoping), `tests/Feature/PublicInvoiceDownloadTest.php` (reference + any phone spelling, wrong phone 404, pending reservation has no invoice, group booking, rate limit), `tests/Feature/TenantTest.php` (payment accounts, tax IDs, branding upload/removal), and `tests/Feature/RateLimitTest.php` (3 tests: login per-IP/per-email, public portal writes per-IP, messaging per-user), `tests/Feature/StaffPinLoginTest.php` (13 tests: PIN login by email and by registration

number, wrong/unset/unknown-identifier rejections, disabled accounts, 4-digit validation, and the set-pin guards — role hierarchy, no self-service, confirmation required).

- Run: `php artisan test` (currently **188 passing, 646 assertions**).

4.11 Realtime (Reverb + Echo)

- **Server:** `laravel/reverb` runs a standalone WebSocket server (`php artisan reverb:start` , default `ws://localhost:8080`). Configure `REVERB_APP_ID` , `REVERB_APP_KEY` , `REVERB_APP_SECRET` , `REVERB_HOST` / `REVERB_PORT` and the public/private ports in `.env` .
- **Client:** the SPA connects with `laravel-echo` (`pusher-js` transport) against `REVERB_*` config, authenticated via the Sanctum bearer token (channel auth goes through `/api/broadcasting/auth`).
- **Channels:** per-user private channel `private-users.{user_id}` (incoming messages, meeting invites, SOS rings), per-tenant `private-tenants.{tenant_id}` (announcements, SOS, handovers, guest messages, staff locations, escalations) and per-thread `private-conversation.{id}` / `private-group.{id}` (message, reply, pin/star, poll and task-convert events scoped to the open thread). Online state is derived from last activity; no presence channels are used.
- **Security:** private channels are authorised by `BroadcastChannel::check` — a user may only subscribe to their own user channel and to channels of tenants they belong to. Sensitive payloads (view-once media, call signaling data) never travel on public channels.

4.12 Rate limiting

Defined as named limiters in `AppServiceProvider::boot()` and applied in `routes/api.php` :

Limiter	Key	Budget	Applied to
<code>api</code>	<code>user {user_id}</code> (authenticated) / IP (guest)	120 / min (authenticated), 30 / min (guest)	whole authenticated group (<code>auth:sanctum</code> → <code>tenant</code> → <code>throttle:api</code>) and <code>/api/broadcasting/auth</code>
<code>auth</code>	IP + email (or the PIN-login identifier)	5 / min	<code>auth/login</code> , <code>auth/login-pin</code> , <code>auth/register</code> — blocks credential stuffing; both sign-in forms share one budget
<code>public</code>	IP	10 / min	portal writes & lookups: <code>public/booking-requisitions</code> , <code>public/reservations</code> , <code>public/payments/initiate</code> , <code>public/booking-requisitions/status</code> , <code>public/invoices/download</code>
<code>messaging</code>	<code>user {user_id}</code>	30 / min	messages, groups, reactions, statuses, calls (<code>level:20</code> group)
<code>webhook</code>	IP	60 / min	<code>payments/clickpesa/callback</code> (server-to-server; generous to avoid breaking payment verification)

Violations return `429 Too Many Requests` with the standard Laravel JSON body (`Retry-After` included). Counter storage uses the configured cache store (database by default); flush `cache` in tests to keep rate-limit tests isolated.

5. Frontend Architecture

5.1 Project layout

```
src/
  api/           axios instance + endpoint groups (index.js, echo.js)
  components/    PhoneInput, CountryCitySelect, StayDates, PaymentMethodSelect,
ChangePasswordForm, ModulePlaceholder
  composables/   useCallManager.js (WebRTC + call ring), useConversation.js, useBroadcast.js
  config/        modules.js – role access matrix
  layouts/       StoreLayout (public + hotel panel), SuperadminLayout
  locales/       i18n.js + en.json + sw.json
  pages/         public/, auth/, dashboards/, reservations/, rooms/, guests/, payments/,
booking/,
                housekeeping/, orders/, menu/, laundry/, fungames/, inventory/, suppliers/,
                procurement/, staff/, reports/, overview/, superadmin/, profile/,
                messages/, statuses/
  router/        index.js – routes + guards
  stores/        auth.js (Pinia, persisted token), session.js (idle watchdog)
  utils/         dates.js, locations.js, payments.js, phone.js
  main.js
```

5.2 API layer (`src/api`)

- `axios.js` : baseUrl from `VITE_API_URL` (default `http://localhost:8000/api`); adds `Authorization: Bearer <auth_token>` from `localStorage`; flattens Laravel pagination (`data.data` + `meta` → top-level); on 401 clears the token and hard-redirects to `/login` .
- `index.js` exposes typed endpoint groups under `/v1`: `publicApi` , `authApi` , `reportApi` , `userApi` , `roomApi` , `guestApi` , `reservationApi` , `paymentApi` , `housekeepingApi` , `inventoryApi` , `supplierApi` , `menuItemApi` , `orderApi` , `laundryApi` , `funGameApi` , `purchaseRequisitionApi` , `purchaseOrderApi` , `goodsReceivedNoteApi` , `bookingRequisitionApi` , `tenantApi` , `superadminReportApi` , `conversationApi` , `groupApi` , `messageActionApi` (react/delete/view-once), `statusApi` (post/list/view/like), `callApi` (start/accept/reject/decline/end/miss), `featuresApi` (reply/pin/star/polls/templates/scheduled/announcements/search/export/translate/escalate/retention/preferences/handovers/guest SMS/nearby/SOS/forward), `roomLinkApi` (index/searchRooms/link/unlink), `taskGroupApi` (store/convert), `meetingApi` (index/store/respond/searchUsers), `sosApi` (index/initiate/acknowledge/resolve).
- `echo.js` sets up the `laravel-echo` instance from `VITE_REVERB_*` env vars, authorises private channels through the SPA's `axios` auth headers, and exports `echo` for the broadcast composables.
- PIN sign-in: `authApi.loginPin({ identifier, pin })` mirrors `authApi.login` (same response shape); `userApi.setPin(id, { pin, pin_confirmation })` lets admins/managers assign staff login PINs from the Staff page.

- `reportApi.overview(params)` sends per-section filters/pagination (`staff_search` , `role` , `status` , `in_house_search` , `upcoming_search` , `housekeeping_status` , `page`); nested sections return raw { `data` , `links` , `meta` } (the interceptor only flattens top-level pagination).

5.3 Routing & access control

- Three areas: public portal (`/` , `/hotels/:id` , `/booking`), hotel panel (`/app/*`), superadmin (`/superadmin/*`).
- `router.beforeEach` : requires `auth_token` for `requiresAuth` ; guest routes bounce signed-in users to their dashboard; fetches `/me` before role/module checks; enforces `role` meta and module access via `canAccess` .
- Access matrix (`src/config/modules.js`): explicit role allow-lists per module with optional extra `permission` gate. Empty `roles` = everyone (dashboard, profile). Superadmin bypasses all lists.

5.4 Stores (Pinia)

- `auth.js` : `token` (persisted via `pinia-plugin-persistedstate` `pick: ['token']`), `user` , `permissions` , `mustChangePassword` ; `ROLE_LEVELS` ; helpers `can()` , `hasPermission()` , `canAccess()` ; actions `login` , `loginPin` (identical session handling to `login` , used by the PIN keypad mode), `logout` , `fetchProfile` , `changePassword` .
- `session.js` : **5-minute idle timeout**; leaving the page (tab switch/minimise/close) logs out **immediately** — `visibilitychange` → `terminate()` , and `pagehide` clears the persisted token synchronously so a closed tab always returns to login; activity listeners; auto-logout → login page.
- `messages.js` (Pinia): conversations + groups merged into one feed, unread counts, active thread state; subscribes to `MessageSent` / `MessageRead` / `MessageDeleted` / `MessageReacted` via Echo to update the UI live.

5.5 i18n

- `vue-i18n` v11; locale files `en.json` and `sw.json` ; default locale from `localStorage` , toggle button in the top bar (`EN` / `SW`).
- Always translate UI strings via `$t('section.key')` ; add new keys to **both** files.

5.6 Key components

Component	Purpose
<code>PhoneInput</code>	Phone input with country-code select, format via <code>libphonenumber-js</code> ; emits <code>update:modelValue</code> + <code>update:countryCode</code> .
<code>CountryCitySelect</code>	Cascading country/city pickers (country-state-city); emits <code>update:countryCode</code> , <code>update:country</code> , <code>update:city</code> .
<code>StayDates</code>	Arrival/departure/days inputs; keeps departure & day count in sync.
<code>PaymentMethodSelect</code>	Method + provider select with contextual notices (selcom auto-pay, mobile-money confirmation).

Component	Purpose
<code>BookingStatusTracker</code>	Public booking status lookup card (reference + phone).
<code>InvoiceDownloadCard</code>	Public guest self-service invoice download (reference + phone) via <code>publicApi.invoiceDownload</code> ; saves the blob with the server-provided filename (<code>saveBlob</code> util).
<code>ChangePasswordForm</code>	Change-password form.
<code>ModulePlaceholder</code>	Generic "under construction" page for unimplemented modules.
<code>ConversationItem</code> / <code>MessageBubble</code>	Conversation row (avatar, last message, unread badge, online/status ring) and message bubble (ticks, media thumbnail, reactions row , long-press/hover context menu for delete / view-once toggle).
<code>ViewOnceMedia</code>	View-once image/video: covers the media, exposes <code>media_url</code> exactly once, then marks <code>is_media_seen</code> via <code>messageActionApi</code> .
<code>ReactionPicker</code>	Emoji picker used to react to a message (toggles on/off).
<code>StatusRing</code>	Avatar ring showing a colleague has an active status; navigates to the Statuses page.
<code>StatusItem</code>	A single 24-hour status (photo/video/text) with viewer + reaction counts.
<code>CallIncomingOverlay</code>	Full-screen incoming-call modal (audio/video) with accept/reject; uses <code>useCallManager</code> .

Key pages: `pages/messages/` (conversation list + thread composer with replies, priority, polls, templates, scheduled send, forwarding, pin/star, translate, search, export, mute/DND, room-link + task groups, and the **Workspace panel** with Announcements / Meetings / Handovers / Guest SMS / Nearby Staff / Escalations / SOS / Scheduled / Starred / Retention tabs + the floating SOS button), `pages/statuses/` (my status + status feed), `pages/calls/` for call history (missed/active logs).

Composables: `useBroadcast` (subscribes to Echo private channels), `useConversation` (thread loading, sending, polling fallback), `useCallManager` (peer connection, `getUserMedia` , ring/answer/end signaling over `CallEvent` broadcasts).

Mobile tables: list tables are wrapped in `<div class="table-scroll">` (`overflow-x: auto`) so only the table scrolls horizontally on small screens; `html` / `body` both have `overflow-x: hidden` and page-level horizontal scrolling should never appear.

5.7 Utilities

- `dates.js` — `todayISO()` , `addDays()` , `daysBetween()` .
- `payments.js` — method/provider constants, `isAutoPaid()` , `requiresConfirmation()` , `requiresProvider()` .
- `locations.js` — country/city lists, `getCountryName()` , `findCountryCode()` .
- `phone.js` — `normalizePhoneNumber()` .

5.8 Frontend testing

- Vitest unit tests in `src/__tests__` .
- Playwright e2e specs in `e2e/` (`login.spec.js` , `modules.spec.js` , `vue.spec.js`).
- Scripts: `npm run build` , `npm run lint` (eslint), `npm run format` (oxfmt), `npm test` .

6. API Reference — Key Endpoints

6.1 Public

POST /api/v1/auth/login-pin

Body: `identifier` (username/email **or** registration number), `pin` (exactly 4 digits).

Response: identical to password login — `200 { message, token, user }` . Unknown identifiers, unset PINs and wrong PINs all return one generic `401 { "message": "Invalid credentials." }` ; throttled 5/min together with `auth/login` and `auth/register` under the shared `auth` limiter.

GET /api/v1/public/hotels

Params: `search` (hotel name/city LIKE), `country` , `city` , `guests` (min occupancy), `check_in` , `check_out` , `per_page` .

Response (paginated `hotels` + `meta`): each hotel includes `tenant_id` , `hotel_name` , `city` , `country` , `available_rooms` , `starting_price` , `room_types` .

GET /api/v1/public/availability

Params: `hotel_id` , `check_in` , `check_out` , `room_type` , `booking_type` , `guests` .

Response: `{ hotel, rooms: [...], payment_methods: [...] }` .

POST /api/v1/public/reservations

Validates guest details + booking (`booking_date` = `after_or_equal:today` + `before_or_equal:check_in_date`). With `room_selections[]` returns `201 { booking_reference, total_amount, message }` ; otherwise a booking requisition.

POST /api/v1/public/payments/initiate

Body: `booking_reference` , `amount` , `payment_method` , `payment_provider` , `phone` .

Response: `{ message, payment: { transaction_reference|clickpesa_reference, ... } }` .

GET /api/v1/public/invoices/download

Guest self-service invoice PDF. Params: `reference` (booking reference), `phone` (matched by trailing digits, so local/international spellings both work).

Returns the generated PDF (`Content-Disposition: attachment; filename=INV-...pdf` , throttled 10/min) or `404` with a generic message when no confirmed reservation matches — never reveals whether the reference exists.

6.2 Authenticated (Bearer token)

Method	URI	Purpose
GET	/api/v1/auth/me	Current user + permissions
POST	/api/v1/auth/logout	Revoke current token
POST	/api/v1/users/{id}/set-pin	Set a staff member's 4-digit login PIN (pin + pin_confirmation ; level 80+, role-guarded, no self-service)
GET	/api/v1/reservations	List (filters: status, booking_type, from, to, search)
POST	/api/v1/reservations	Create reservation
POST	/api/v1/reservations/{id}/check-in	Check in
POST	/api/v1/reservations/{id}/check-out	Check out
POST	/api/v1/reservations/{id}/cancel	Cancel
POST	/api/v1/reservations/{id}/no-show	Mark no-show
POST	/api/v1/payments	Record a payment
GET	/api/v1/rooms , /api/v1/guests , /api/v1/payments	Module lists
GET	/api/v1/messages/conversations	Direct conversations (merge with groups on the client)
POST	/api/v1/messages/conversations	Start a conversation (scope: hotel global , user_id)
GET	/api/v1/messages/conversations/{id}/messages	Messages + poll (marks them delivered)
POST	/api/v1/messages/conversations/{id}/messages	Send (body and/or media , optional type)
POST	/api/v1/messages/conversations/{id}/read	Mark thread read
GET	/api/v1/messages/unread-count	Direct + group unread total for the nav badge
GET/ POST	/api/v1/messages/groups	List / create group (name , scope , member_ids[])
GET	/api/v1/messages/groups/{id}/messages	Group messages (mark delivered)

Method	URI	Purpose
POST	/api/v1/messages/groups/{id}/messages	Send group message (body / media)
POST	/api/v1/messages/groups/{id}/read	Mark group read (last_read_at)
POST	/api/v1/messages/groups/{id}/members	Add members (user_ids[])
DELETE	/api/v1/messages/groups/{id}/members/{userId}	Remove member (leave when self)
POST	/api/v1/messages/conversations/{convId}/messages/{msgId}/delete	Delete a message (delete_for: me\ everyone) — both parties vanish from threads; everyone removes it everywhere
POST	/api/v1/messages/conversations/{convId}/messages/{msgId}/react	Toggle an emoji reaction (emoji)
GET	/api/v1/messages/conversations/{convId}/messages/{msgId}/media	Consume a view-once media URL (one hit only, then locked)
POST	/api/v1/messages/groups/{groupId}/messages/{msgId}/delete	Delete a group message (delete_for: me\ everyone)
POST	/api/v1/messages/groups/{groupId}/messages/{msgId}/react	Toggle a reaction on a group message
GET	/api/v1/messages/statuses	Status feed (active, < 24 h)
POST	/api/v1/messages/statuses	Post a status (type: text\ photo\ video , body / media)
GET	/api/v1/messages/statuses/{id}	View a status (marks it seen)
POST	/api/v1/messages/statuses/{id}/like	Toggle a reaction/like on a status
POST	/api/v1/messages/calls/start	Start a WebRTC call (type , recipient_id) — rings the callee
POST	/api/v1/messages/calls/{id}/accept / reject / decline / end	Call lifecycle (decline = busy)
POST	/api/v1/messages/calls/{id}/miss	Mark a call missed (ring timeout / auto-end)

Method	URI	Purpose
GET	/api/v1/messages/calls	Call history (missed/active/ended, scoped to caller)
POST	/api/v1/messages/pin / unpin / star / unstar	Pin/star a message (message_type , message_id)
GET	/api/v1/messages/pinned , / api/v1/messages/starred	Your pinned / starred messages
POST	/api/v1/messages/polls/vote	Vote on a poll (poll_id , poll_option_id)
GET/ POST	/api/v1/messages/templates	List / create message templates; PUT / DELETE per id
GET/ POST	/api/v1/messages/scheduled	List / schedule a message (recipient_type , recipient_id , body , send_at , recurrence)
DELETE	/api/v1/messages/scheduled/ {id}	Cancel a scheduled message
GET/ POST	/api/v1/messages/ announcements	List / post an announcement; POST {id}/acknowledge
GET	/api/v1/messages/search	Keyword search across accessible chats (search , chat_type / chat_id , from / to)
GET	/api/v1/messages/export	CSV export of a chat history (chat_type , chat_id)
POST	/api/v1/messages/translate	Translate a body (offline EN↔SW glossary)
GET/ POST	/api/v1/messages/escalations	List / raise an escalation (message_type , message_id , reason)
POST	/api/v1/messages/escalations/ {id}/resolve	Resolve an escalation (resolution , handled_by)
GET/ POST	/api/v1/messages/retention- policies	List / create a retention policy (scope , days , enabled); DELETE per id
GET/ POST	/api/v1/messages/preferences	List / create notification preferences (scope , target_id , muted , muted_until , dnd_enabled , dnd_from , dnd_to , push_enabled)
PUT/ DELETE	/api/v1/messages/preferences/ {id}	Update / delete a preference
GET/ POST	/api/v1/messages/handovers	List / post a shift handover (title , notes , shift_from , shift_to); POST {id}/acknowledge
GET	/api/v1/messages/ rooms , /api/v1/messages/rooms/ search	Linked rooms / search rooms to link (search)
POST	/api/v1/messages/rooms	Link a chat to a room (chat_type , chat_id , room_id)

Method	URI	Purpose
DELETE	/api/v1/messages/rooms/{id}	Unlink a room from a chat
POST	/api/v1/messages/task-groups	Create a task group (room-linked group for housekeeping)
POST	/api/v1/messages/groups/{groupId}/messages/{messageId}/convert-task	Convert a room-linked message into a HousekeepingTask (task_type)
POST	/api/v1/messages/forward	Forward a message (message_type , message_id , target_type , target_id)
PUT/GET	/api/v1/messages/location , /api/v1/messages/nearby	Update your zone/floor; list colleagues updated in the last 30 min (zone , floor)
GET/POST	/api/v1/messages/guest-messages	List / send an outbound guest SMS (phone , body)
GET/POST	/api/v1/messages/meetings	List / schedule a meeting (title , start_at , duration_minutes , conference_type , invitee_ids)
GET	/api/v1/messages/meetings/users	Search staff to invite (search)
POST	/api/v1/messages/meetings/{id}/respond	Accept/decline a meeting invite (response)
GET/POST	/api/v1/messages/sos	List / initiate an SOS alert (message)
POST	/api/v1/messages/sos/{id}/acknowledge / resolve	Acknowledge / resolve an SOS alert
GET	/api/v1/reports/overview	Admin overview with per-section filters + 15/page pagination
GET	/api/v1/owner/dashboard	Owner: consolidated stats + per-hotel comparison across owned hotels
GET	/api/v1/owner/hotels , /api/v1/owner/hotels/{id}	Owner: owned-hotel list / drill-down (404 for unowned)
GET/POST	/api/v1/owners	Superadmin: list / create owner accounts
GET	/api/v1/invoices	List invoices (tenant scoped)
POST	/api/v1/reservations/{id}/invoice	Generate/refresh the folio invoice
GET	/api/v1/invoices/{id}/download	Staff invoice PDF download

Method	URI	Purpose
PUT	/api/v1/tenants/{id}	Superadmin: update hotel details incl. <code>tin</code> , <code>vrn</code> , <code>payment_methods</code> , <code>payment_accounts</code>
POST	/api/v1/tenants/{id}/branding	Superadmin: upload/remove invoice signature & stamp images (multipart <code>signature</code> / <code>stamp</code> , or <code>remove_signature</code> / <code>remove_stamp</code>)

Payment methods payload (superadmin):

```
{ "payment_methods": ["cash", "mobile_money", "bank", "selcom", "card"] }
```

7. Payment semantics

Method	Provider required	Initial status	Confirmation
cash / card	—	<code>completed</code>	immediate
mobile_money	yes (airtel_money, mixx_by_yas, halopesa, mpesa)	<code>pending (online)</code> / <code>awaiting_confirmation (desk)</code>	ClickPesa webhook → <code>completed</code> , or receptionist confirms manually
bank	yes (crdb, nmb, nbc, other)	<code>awaiting_confirmation</code>	hotel staff marks paid
selcom	—	<code>completed</code>	immediate + confirms booking

A mobile-money payment is always **confirmable**: `Payment::isConfirmable()` accepts both `pending` and `awaiting_confirmation` , so the receptionist can verify and confirm an online payment on the Payments page (`payments/{id}/confirm`) using the reference from the guest's SMS — the webhook is a fast-path auto-confirm, not the only path.

`PaymentOptions::defaultMethods()` = all staff methods **except** selcom (shipped disabled). The superadmin can enable it per tenant.

8. Messaging semantics

Delivery / read ticks. A message is `delivered_at` the moment the recipient's device pulls it — any read of the conversation list, the thread, or a `messages` poll marks outgoing messages as delivered. `markRead` sets `read_at` on the conversation (or `last_read_at` on the group membership) when the thread is opened.

- Direct messages: single tick = delivered, double tick = read.

- Group messages: a message is "read" when `read_by_count > 0` (computed in `GroupMessageResource` from loaded members whose `last_read_at >= message created_at`). The UI shows "Seen by N".

Scoping. `scope=hotel` conversations are restricted to the caller's tenant; `scope=global` conversations cross hotels but never expose a superadmin. Group creation enforces the same rule per scope.

Media. `POST .../messages` accepts `body`, `media` (file, ≤ 10 MB) and optional `type`. `type` is inferred from the MIME (`audio/*` → audio, `image/*` → image, else file) when not supplied. `body` is nullable, so audio/attachment-only messages are valid. Files are stored on the public disk under `messages/conversations/{id}` or `messages/groups/{id}` and exposed via `media_url` on the message resource.

View-once media. Messages sent with `view_once: true` are rendered in the client as a covered tile. The recipient taps **once** to reveal; the client calls `GET .../messages/{msgId}/media`, which serves the media and sets `is_media_seen = true`. Subsequent requests return 403 and the UI shows an "opened" state — the media can never be replayed. View-once is only applied to `image / video` types; plain text cannot be view-once.

Deleting messages. `POST .../delete` takes `delete_for: me|everyone`. - Direct conversations: `me` hides the message for the caller only (recipient keeps it); `everyone` removes it for **both** parties. The other side learns instantly via the `MessageDeleted` broadcast (which carries `channel_id` so open threads update live without a refetch). - Group conversations: `me` hides it for the caller; `everyone` soft-deletes it for all current members (restored conversation history still omits it). Group message deletion also broadcasts `MessageDeleted`.

Reactions. `POST .../react` with `emoji` toggles an emoji on a message. Reactions are stored as a JSON map on the message row (`reactions` column keyed by `user_id` → `emoji`), summed per emoji in the message resource. The UI renders the reaction row beneath the bubble (tapping your own reaction removes it). `MessageReacted` broadcasts keep open threads in sync.

Mentions. Typing `@` in the composer surfaces a colleague picker (from `messages/users`). Selecting someone inserts `@FirstName`, sends `mention_user_ids[]` with the message, and the mentioned user gets a highlighted bubble + notification in their thread (via `MessageSent` with `mentions`). Mentions have no tenant-crossing behaviour beyond the existing `scope` rules.

Statuses. Every user can post a **status** (`type: text|photo|video`) that lives for **24 hours** then expires (soft-deleted by the feed query, no cron needed). A status is scoped to the poster's tenant by default (`scope=hotel`) or `scope=global`. The **Statuses page** shows "Your status" plus a feed; viewing a status records a `StatusView` (deduplicated per user), and `POST /statuses/{id}/like` toggles an emoji reaction (`StatusReaction`). In the conversation list, a colleague's avatar shows a **status ring** when they have an active status — fed live by the `StatusPosted` broadcast on the tenant channel.

Calls (WebRTC). The `CallController` orchestrates the signaling: 1. `POST /calls/start` (`type: audio|video`, `recipient_id`) creates a `Call` (`ringing`), broadcasts `CallStarted` to `private-users.{recipient_id}`, and records a `CallEvent` (`ring`). 2. The callee's `CallIncomingOverlay` offers accept/reject. Accepting calls `POST /calls/{id}/accept` (`status` → `active`), broadcasts `CallEvent` (`accept`) so the initiator opens the peer connection. Rejecting → `rejected`; a busy callee → `decline` (`declined`). 3. Media is exchanged peer-to-peer over WebRTC (no server-side media). Call state rides the `CallEvent` broadcasts; `end` (either side) →

ended , ring timeout → missed . 4. Call history (GET /calls) lists the caller's own calls (missed/active/ended) with callers/callees, type and timestamps.

Realtime. All of the above are pushed over Reverb/Echo (see §4.11): MessageSent , MessageRead , MessageDeleted , MessageReacted , StatusPosted , CallStarted , CallEvent , MemberJoined / MemberLeft . The SPA listens on private-users.{id} (direct) and private-tenants.{tenant_id} (group/global). A polling fallback remains for slow WebSocket connections (unread count + thread refetch).

The 24-feature enhancement set. Everything below lives under messages/* , is level-20+ (any employee) unless noted, and rides the private channels above.

- **Replies.** POST .../messages accepts reply_to_id . The message row self-joins (reply_to), the resource includes reply_to: { message_id, sender_id, sender_name, body } , and ReplySent broadcasts the full reply (conversation → MessageResource , group → GroupMessageResource). The composer shows a **reply bar** while composing; tapping a quoted reply scrolls to the original message.
- **Priority.** POST .../messages accepts priority: normal|urgent . Urgent messages get a red border, an "urgent" tag, a pin **and** are auto-escalated when unread past the command threshold (messaging:escalate).
- **Polls.** Sending a poll: { question, multiple, options[] } creates Poll / PollOption rows and broadcasts PollCreated . Voting is POST messages/polls/vote (poll_id , poll_option_id), once per user (unique poll_votes), tally + percentages recomputed and broadcast via PollVoted . Messages with a poll render the live option bars inline in the thread.
- **Pin & star.** PinStarController upserts PinnedMessage / StarredMessage (per user), broadcasts MessagePinned / Unpinned / Starred / Unstarred on the thread channel so open chats update live, and exposes GET messages/pinned / messages/starred (each returns a copy of the message at pin time, so later edits/deletes don't corrupt the pinned snapshot).
- **Templates.** MessageTemplate is a personal/tenant snippet store (name , body , category , is_global). The composer's template picker inserts the body directly.
- **Scheduled messages.** ScheduledMessage holds recipient_type (conversation/group), recipient_id , body / media , send_at and optional recurrence . The messaging:send-scheduled command (runs every minute) dispatches due sends through the normal send pipeline (so MessageSent still fires), then stamps last_sent_at ; repeating schedules are re-queued, one-off sends are soft-cancelled.
- **Forwarding.** POST messages/forward (message_type , message_id , target_type , target_id) copies the source message into the target chat with forwarded_from_type / forwarded_from_id provenance and a "Forwarded" label in the UI. Source and target access are both validated (participant/member + same-tenant), deleted and view-once messages cannot be forwarded, and the copy dispatches MessageSent / GroupMessageSent on the target thread.
- **Announcements.** Announcement posts tenant-wide and broadcasts AnnouncementPosted ; every recipient sees it in the **Workspace → Announcements** tab and POST {id}/acknowledge records an AnnouncementAcknowledgement (broadcast AnnouncementAcknowledged) so the poster sees "N acknowledged".
- **Search & export.** GET messages/search (keyword + optional chat_type / chat_id , from / to) matches ConversationMessage and GroupConversationMessage that the caller can access and returns them newest-first with a matching snippet. GET messages/export streams a CSV of a chat's history (id, sender, body, type, timestamps).

- **Translate.** `POST messages/translate` with `text` returns the offline `TranslateService` result — a bundled EN↔SW glossary with direct + reverse lookups (no external API, works offline). The bubble's translate button toggles original / translated text.
- **Escalations.** `MessageEscalation` is polymorphic (`message_type` , `message_id`) with `escalated_by` , optional `handled_by` / `handled_at` . `POST messages/escalations` raises it (broadcast `MessageEscalated` to the tenant channel); managers see a tab in Workspace and `POST {id}/resolve` marks it handled.
- **Retention.** `RetentionPolicy` (`scope` = `global/conversation/group` , `days` , `enabled`) is set by admins (the Workspace Retention tab is admin-only). `messaging:purge-expired` soft-deletes messages older than the policy; expired scheduled messages are also cleaned up.
- **Notification preferences / DND.** `NotificationPreference` is polymorphic (`scope` = `conversation/group/global` , `target_id`): mute a chat (with optional `muted_until`) or set a daily DND window (`dnd_from` / `dnd_to`). The muted chat shows a speaker-off icon; `push_enabled` gates push-style notifications.
- **Shift handovers.** `ShiftHandover` captures `title` , required `notes` , `shift_from` / `shift_to` . `POST messages/handovers` broadcasts `ShiftHandoverPosted` ; `POST {id}/acknowledge` records who acked and broadcasts `ShiftHandoverAcknowledged` .
- **Room-linked chats & task groups.** `ConversationRoom` binds a chat to a `Hotel\Room` . `TaskGroupController::store` creates a **task group** (group chat with `is_task_group` , a `task_type` , and a linked room); its members are room-relevant staff. `POST .../convert-task` turns a room-linked message into a `HousekeepingTask` (broadcast `TaskConverted`), so a "Room 401 needs sheets" chat message becomes an actual housekeeping assignment.
- **Staff presence.** `PUT messages/location` upserts `StaffLocation` (`zone` , `floor` , `updated_at`); `GET messages/nearby` lists colleagues updated in the last 30 minutes, filterable by zone/floor. Updates broadcast `StaffLocationUpdated` on the tenant channel so the Workspace nearby tab refreshes live.
- **Guest SMS bridge.** `POST messages/guest-messages` (`phone` , `body` , optional `template_id`) creates an outbound `GuestMessage` (direction `outbound`) and broadcasts `GuestMessagePosted` ; history is tenant-scoped.
- **Meetings.** `Meeting` (`start_at` , `duration_minutes` , `conference_type` , `status`) with `MeetingInvitee` rows. Scheduling broadcasts `MeetingInvited` to each invitee's user channel; `POST {id}/respond` (`accepted` / `declined`) broadcasts `MeetingResponseChanged` so the organizer's list updates live (the `index` response includes `my_invitee` per meeting).
- **SOS alerts.** `SosController::initiate` (`message`) broadcasts `SosAlertInitiated` to the tenant channel with a prominent pulse in the UI; any staff member acknowledges (`ack_count` , `ack_user_ids`) or resolves (`SosAlertResolved`). The SOS button floats over the chat list, including on the login-protected hotel panel.

9. Deployment notes

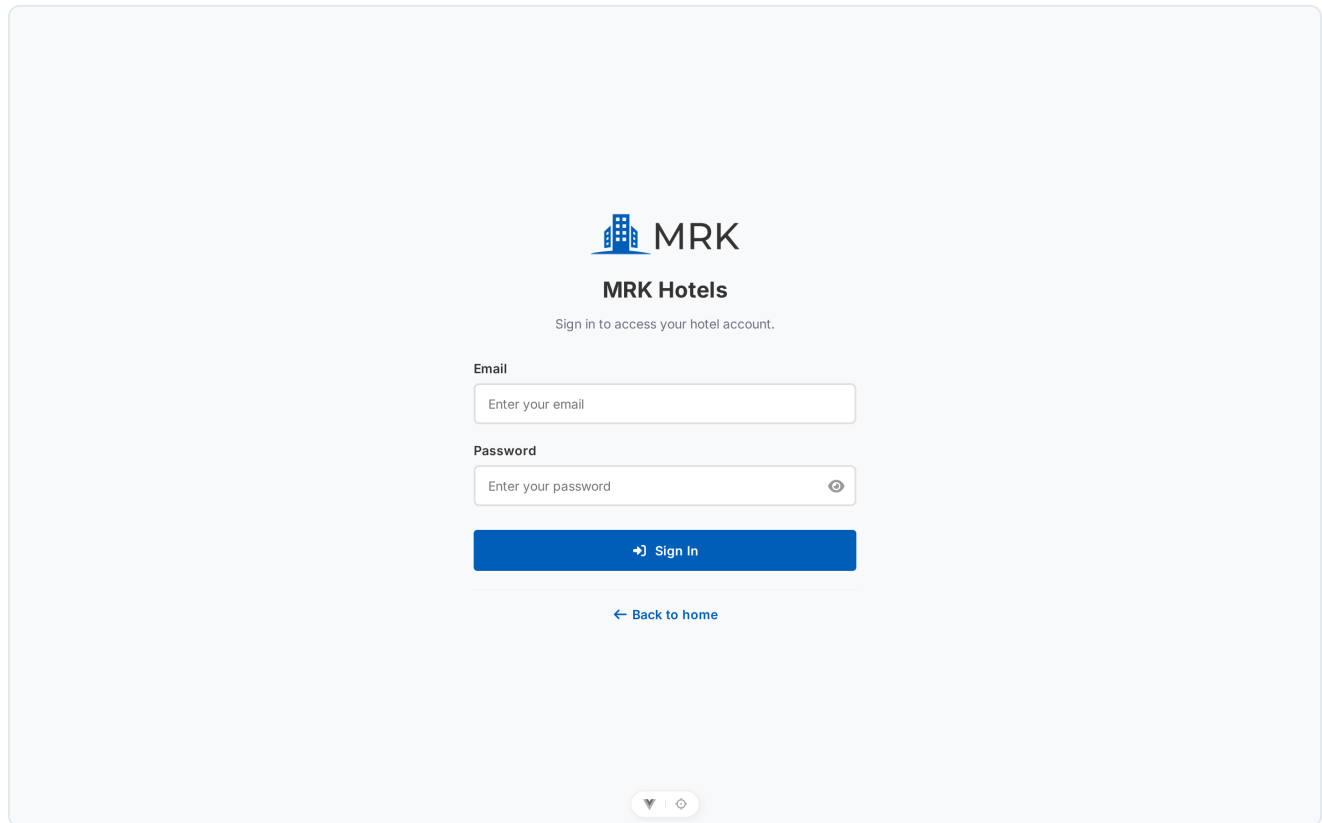
- Set `APP_ENV` , `APP_DEBUG` , `APP_URL` , `SANCTUM_STATEFUL_DOMAINS` (add your frontend host) and `CLICKPESA_*` keys (sandbox live URL, service fee percent).

- `APP_URL` must be the public API origin (including port) — tenant branding image URLs (`signature_url` / `stamp_url`) are built from it.
 - Run `php artisan storage:link` so uploaded branding images under `storage/app/public/tenants/{id}` are reachable.
 - CORS is configured in `config/cors.php` : `Content-Disposition` is exposed so the SPA can read the generated invoice filename from download responses; keep it exposed when hardening origins.
 - Build the frontend with `npm run build` and serve `dist/` behind the same-origin as `VITE_API_URL` or configure CORS.
 - Run `php artisan migrate --force` in production; schedule nothing currently (no queued jobs required by the current flow — ClickPesa is handled synchronously + webhook).
 - Realtime in production: run `php artisan reverb:start` behind a process supervisor (or Reverb's own fleet config), set `REVERB_*` env vars on both apps, and make sure the SPA's `VITE_REVERB_*` values match. If the WebSocket handshake is proxied, also expose `/api/broadcasting/auth` through your API host.
 - Tests: `php artisan test` (SQLite in-memory) and `npm test` / Playwright for the frontend.
-

10. Screens

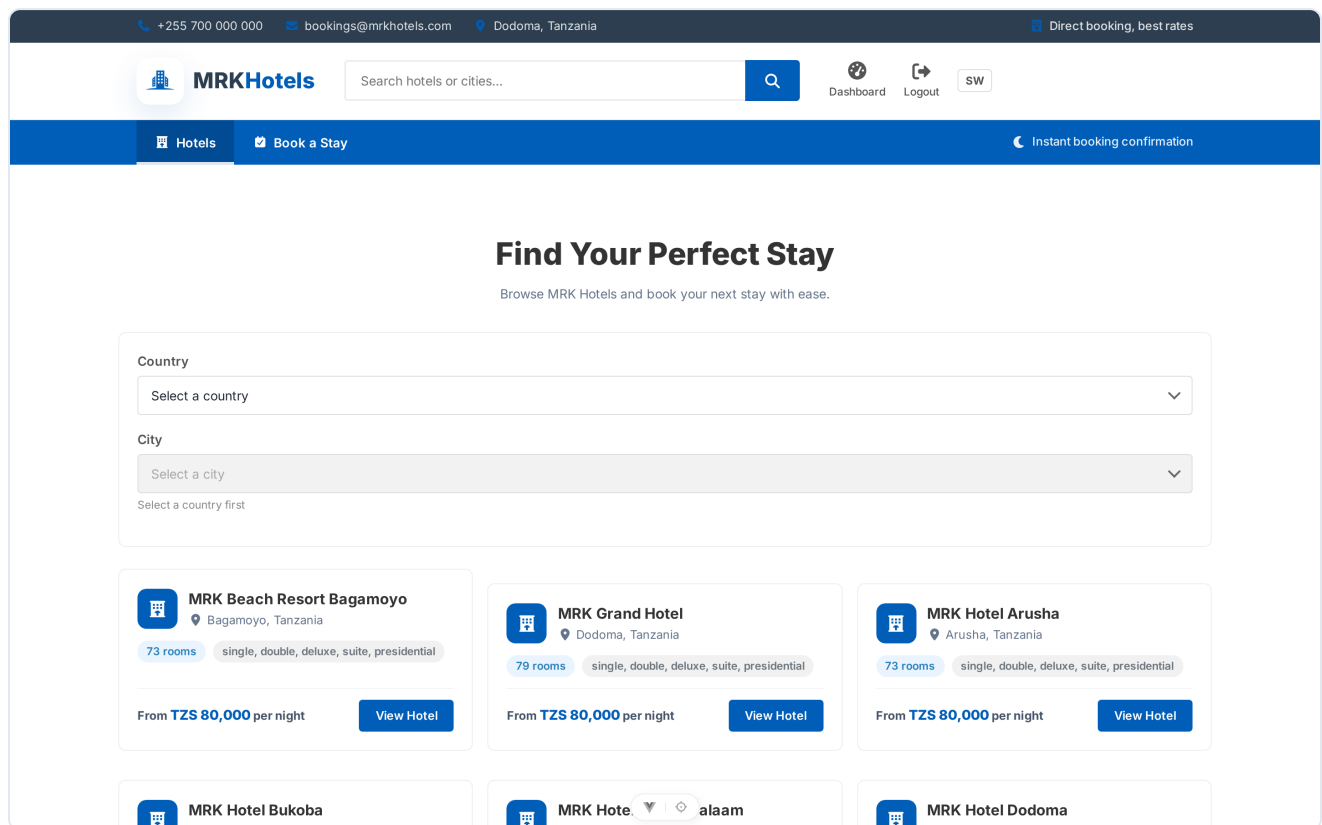
Captured from the running demo (MRK Grand Hotel + the platform superadmin). Public and hotel-panel screens are shared with the user manuals; the superadmin screens appear only here.

10.1 Public portal and sign-in



The image shows the MRK Hotels sign-in page. At the top, there is the MRK Hotels logo and the text "Sign in to access your hotel account." Below this, there are two input fields: "Email" with the placeholder "Enter your email" and "Password" with the placeholder "Enter your password" and an eye icon for toggling visibility. A blue "Sign In" button is positioned below the password field. A link "← Back to home" is located below the button. At the bottom center, there is a small dropdown menu with a downward arrow and a circular refresh icon.

Sign-in page (`/login`) — dual-mode: email + password, or username/registration number + 4-digit PIN on the iPOS-style keypad.



The image shows the MRK Hotels public booking portal home page. The header includes contact information (+255 700 000 000, bookings@mrkhotels.com, Dodoma, Tanzania), a search bar, and links to Dashboard, Logout, and SW. The main navigation bar has "Hotels" and "Book a Stay" tabs, with "Instant booking confirmation" on the right. The main content area is titled "Find Your Perfect Stay" with the subtitle "Browse MRK Hotels and book your next stay with ease." Below this, there are filters for "Country" and "City". The main content area displays a grid of hotel cards. Each card shows the hotel name, location, number of rooms, room types, and a "View Hotel" button. The hotels listed are MRK Beach Resort Bagamoyo, MRK Grand Hotel, MRK Hotel Arusha, MRK Hotel Bukoba, MRK Hotel ... (partially visible), and MRK Hotel Dodoma.

Public booking portal home (`/`) — hotel directory, country/city filters, invoice download.

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

Direct booking, best rates

MRKHotels

DashboardLogoutSW

HotelsBook a StayInstant booking confirmation

Back to Hotels

MRK Beach Resort Bagamoyo

Beach Road, Indian Ocean, Bagamoyo, Tanzania

0755123456

Book Now

Rooms (73)

available

Room 101

single · Floor 1

Up to 2 guests

TZS 80,000 per night

available

Room 102

single · Floor 1

Up to 2 guests

TZS 80,000 per night

available

Room 103

single · Floor 1

Up to 2 guests

TZS 80,000 per night

available

Room 104

single · Floor 1

Up to 2 guests

TZS 80,000 per night

cleaning

Room 105

single · Floor 1

Up to 2 guests

TZS 80,000 per night

available

Room 106

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 107

double · Floor 1

Up to 2 guests

TZS 120,000 per night

occupied

Room 108

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 109

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 110

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 111

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 112

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 113

double · Floor 1

Up to 2 guests

TZS 120,000 per night

available

Room 114

deluxe · Floor 1

Up to 3 guests

TZS 180,000 per night

available

Room 115

deluxe · Floor 1

Up to 3 guests

TZS 180,000 per night

available

Room 116

deluxe · Floor 1

Up to 3 guests

TZS 180,000 per night

available

Room 117

deluxe · Floor 1

Up to 3 guests

TZS 180,000 per night

available

Room 118

suite · Floor 1

Up to 4 guests

TZS 300,000 per night

available

Room 201

single · Floor 2

Up to 2 guests

TZS 80,000 per night

available

Room 202

single · Floor 2

Up to 2 guests

TZS 80,000 per night

available

Room 203

single · Floor 2

Up to 2 guests

TZS 80,000 per night

available

Room 204

single · Floor 2

Up to 2 guests

TZS 80,000 per night

available

Room 205

single · Floor 2

Up to 2 guests

TZS 80,000 per night

cleaning

Room 206

double · Floor 2

Up to 2 guests

TZS 120,000 per night

available

Room 207

double · Floor 2

Up to 2 guests

TZS 120,000 per night

available

Room 208

double · Floor 2

Up to 2 guests

TZS 120,000 per night

available

Room 209

double · Floor 2

Up to 2 guests

TZS 120,000 per night

cleaning

Room 210

double · Floor 2

Up to 2 guests

TZS 120,000 per night

maintenance

Room 211

double · Floor 2

Up to 2 guests

TZS 120,000 per night

available

Room 212

double · Floor 2

Up to 2 guests

TZS 120,000 per night

cleaning

Room 213

double · Floor 2

Up to 2 guests

TZS 120,000 per night

available

Room 214

deluxe · Floor 2

Up to 3 guests

TZS 180,000 per night

occupied

Room 215

deluxe · Floor 2

Up to 3 guests

TZS 180,000 per night

occupied

Room 216

deluxe · Floor 2

Up to 3 guests

TZS 180,000 per night

maintenance

Room 217

deluxe · Floor 2

Up to 3 guests

TZS 180,000 per night

available

Room 218

suite · Floor 2

Up to 4 guests

TZS 300,000 per night

available

Room 301

single · Floor 3

Up to 2 guests

TZS 80,000 per night

occupied

Room 302

single · Floor 3

Up to 2 guests

TZS 80,000 per night

occupied

Room 303

single · Floor 3

Up to 2 guests

TZS 80,000 per night

occupied

Room 304

single · Floor 3

Up to 2 guests

TZS 80,000 per night

Page 29 of 43

Public hotel detail (`/hotels/{id}`) — rooms, rates, occupancy.

10.2 Hotel panel

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Welcome back, overview of your hotel

Here is what is happening at MRK Grand Hotel today.

Hotel Admin

0

GUESTS IN HOUSE

0

CHECK-INS TODAY

0

CHECK-OUTS TODAY

0

UPCOMING RESERVATIONS

TZS 0

REVENUE TODAY

10.1%

OCCUPANCY RATE

0

OPEN ISSUE REPORTS

Room Status

View rooms →

Total79

Available66

Occupied8

Cleaning2

Alerts

0 pending purchase requisitions

0 low/out-of-stock inventory items

TZS 0 in pending payments

Staff dashboard (`/app`).

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Reservations

Bookings, check-ins and check-outs.

Refresh

Active

Checked Out

Cancelled

Status

All

Booking type

All

From

mm/dd/yyyy

To

mm/dd/yyyy

Search

Guest or reference...

Clear

GUEST	TYPE	ROOM	STAY	TIMES	TOTAL	BALANCE	STATUS	ACTIONS
No reservations found.								

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

Hotels

Book a Stay

Sign In

Support

Help Centre

Privacy Policy

Terms & Conditions

Contact

+255 700 000 000

bookings@mrkhotels.com

Dodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Reservations.

Page 30 of 43

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Rooms

Manage rooms, rates and their current status.

Status

Room Type

Search

All

All

Room number...

Clear

ROOM	TYPE	FLOOR	RATE / NIGHT	CAPACITY	STATUS	ACTIONS
101	Single	Floor 1	TZS 80,000	2	available	EditStatus
102	Single	Floor 1	TZS 80,000	2	available	EditStatus
103	Single	Floor 1	TZS 80,000	2	available	EditStatus
104	Single	Floor 1	TZS 80,000	2	available	EditStatus
105	Single	Floor 1	TZS 80,000	2	available	EditStatus
106	Double	Floor 1	TZS 120,000	2	occupied	Edit
107	Double	Floor 1	TZS 120,000	2	available	EditStatus

Rooms.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Guests

Guest profiles captured at booking and check-in.

Search

VIP

Nationality

Name, email or phone...

All

e.g. Tanzania

Clear

GUEST	CONTACT	ID	NATIONALITY	TYPE	ACTIONS
victor zachy	victor@gmail.com +255683870268	·	Tanzania	Regular	
Adam Abbas	adam-abbas1@gmail.com 07700000001	·	Tanzanian	Regular	
Agnes Abbas	agnes-abbas2@gmail.com 07700000002	·	Tanzanian	Regular	
Aisha Abbas	aisha-abbas3@gmail.com 07700000003	·	Tanzanian	Regular	
Ally Abbas	ally-abbas4@gmail.com 07700000004	·	Tanzanian	Regular	
Amani Abbas	amani-abbas5@gmail.com 07700000005	·	Tanzanian	Regular	
Amina Abbas	amina-abbas6@gmail.com	·	Tanzanian	Regular	

Guests.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Payments

Payments against reservations and F&B orders.

Refresh

StatusAllMethodAllFrommm/dd/yyyyTo mm/dd/yyyyClear

PAYMENT	RESERVATION	PAYER	AMOUNT	METHOD	STATUS	ACTIONS
PAY-2026-0001 2026-08-09 07:47	victor zachy - room -	+255683870268	TZS 420,000	Cash	completed	Invoice

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Payments.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Booking Requests

Enquiries from the public booking portal.

Refresh

StatusAllBooking typeAllSearchName or reference...Clear

REFERENCE	GUEST	STAY	TYPE	DETAILS	STATUS	ACTIONS
No booking requests found.						

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Booking requisitions.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Housekeeping

Cleaning and room maintenance tasks.

Refresh

Status

House status

Room status

Room

All

All

All

All rooms

Clear

ROOM

GUESTS (PAX)

HOUSE STATUS

ASSIGNED TO

ROOM STATUS

ARRIVAL (DATE & TIME)

DEPARTURE · NIGHTS

STATUS

ACTIONS

No tasks found.

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

Hotels

Book a Stay

Sign In

Support

Help Centre

Privacy Policy

Terms & Conditions

Contact

+255 700 000 000

bookings@mrkhotels.com

Dodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Housekeeping.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Menu Items

Restaurant and bar menu management.

Refresh

Department

Category

Available

Search

All

e.g. main course

All

Item name...

Clear

ITEM

CATEGORY

DEPARTMENT

PRICE

COST

MARGIN

STATUS

ACTIONS

Banana Fritters (Zena)

Sweet banana fritters with honey

Desserts

Restaurant

TZS 4,000

TZS 1,600

60%

Available

-

Beef Mchuzi na Wali

Beef stew served with steamed rice

Main Courses

Restaurant

TZS 11,000

TZS 4,800

56%

Available

-

Beef Mishkaki (Skewers)

Marinated grilled beef skewers

Grills

Restaurant

TZS 14,000

TZS 6,500

54%

Available

-

Beef Nyama Choma (500g)

Char-grilled beef, served with kachumbari

Grills

Restaurant

TZS 25,000

TZS 12,000

52%

Available

-

Beef Pilau

Spiced rice with beef, a coastal classic

Main Courses

Restaurant

TZS 12,000

TZS 5,200

57%

Available

-

Beef Samosa (2 pcs)

Crispy pastry filled with spiced beef

Appetizers

Restaurant

TZS 4,000

TZS 1,500

63%

Available

-

Black & White Whisky (50ml)

Drinks

Bar

TZS 7,000

TZS 3,000

57%

Available

-

Menu items.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

F&B Orders

Restaurant and bar orders.

Refresh

DepartmentStatusOrder typePaymentDate

AllAllAllAllmm/dd/yyyyClear

ORDERLOCATIONGUEST / TABLEWAITERORDER TYPEITEMSTOTALSTATUSPAYMENTACTIONS

No orders found.

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

F&B orders.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Laundry

Wash, iron and dry-clean orders from guests.

Refresh

ServiceStatusPaymentOrder dateSearch

AllAllAllmm/dd/yyyyOrder No.Clear

ORDER NO.GUESTROOMSERVICETOTAL CHARGEORDER DATEACTIONS

No laundry orders found.

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Laundry.

Page 34 of 43

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Fun & Games

Gaming sessions booked with an attendant supervisor.

Refresh

StatusBooking dateGaming attendantSearch

Allmm/dd/yyyyAllGame nameClear

ORDER NO.	GAME	GUEST	GAMING ATTENDANT	BOOKING DATE	ARRIVAL TIME	TOTAL CHARGE	STATUS	ACTIONS
No gaming orders found.								

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

▼

◇

Fun & games.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Inventory

Stock items and stock movements.

Refresh

CategoryStatusSearchLow stock only

AllAllItem name...NoClear

ITEM	CATEGORY	STOCK	REORDER	UNIT COST	SUPPLIER	STATUS	ACTIONS
No inventory items found.							

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

▼

◇

Inventory.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Suppliers

Manage supplier information.

Refresh

StatusAll

CategoryAll

SearchSupplier name...Clear

SUPPLIER	CONTACT	CATEGORY	TERMS	BALANCE	RATING	STATUS	ACTIONS
No suppliers found.							

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Suppliers.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Purchase Requisitions

Purchase requests raised by hotel departments.

Refresh

StatusAll

PriorityAll

Departmente.g. housekeepingClear

REFERENCE	DEPARTMENT	REQUESTED BY	ITEMS	TOTAL	PRIORITY	STATUS	ACTIONS
No requisitions found.							

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

HotelsBook a StaySign In

Support

Help CentrePrivacy PolicyTerms & Conditions

Contact

+255 700 000 000bookings@mrkhotels.comDodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Purchase requisitions.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Purchase Orders

Manage purchase orders placed with suppliers.

Search

Status

Supplier

Reference or supplier...

All

All suppliers

Clear

REFERENCE	SUPPLIER	ITEMS	TOTAL	DELIVERY	STATUS	ACTIONS
No purchase orders found.						

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

Hotels

Book a Stay

Sign In

Support

Help Centre

Privacy Policy

Terms & Conditions

Contact

+255 700 000 000

bookings@mrkhotels.com

Dodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Purchase orders.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Goods Received Notes

Record incoming goods deliveries.

PO

Inspection status

From

To

All POs

All

mm/dd/yyyy

mm/dd/yyyy

Clear

REFERENCE	PO	SUPPLIER	RECEIVED	DELIVERY NOTE	INSPECTION	ACTIONS
No goods received notes found.						

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

Hotels

Book a Stay

Sign In

Support

Help Centre

Privacy Policy

Terms & Conditions

Contact

+255 700 000 000

bookings@mrkhotels.com

Dodoma, Tanzania

© 2026 MRK Hotels. All rights reserved.

Goods received notes.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Staff Management

Manage hotel staff accounts and roles.

Refresh

New Staff

Role

Department

Search

Status

All

All

Name, email or registration no...

All

Clear

STAFF	CONTACT	ROLE	DEPARTMENT	POSITION	LAST LOGIN	STATUS	ACTIONS
Adam Liwenga -	adam-liwenga@mrk-grand.mrkhotels.test 07500000001	Manager	Administration	-	Never	Active	Edit Invite Deactivate
Manager EMP-2026-0002	manager@mrkhotels.test -	Manager	Administration	-	2026-08-09 17:00	Active	Edit Invite Deactivate

MRKHotels

MRK Hotels connects travellers with comfortable stays across Tanzania — book directly for the best rates.

Quick Links

Hotels

Book a Stay

Sign In

Support

Help Centre

Privacy Policy

Terms & Conditions

Contact

+255 700 000 000

bookings@mrkhotels.com

Dodoma, Tanzania

Staff.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Messages

Private chats with your team and across the MRK Hotels network.

Workspace

Refresh

New Message

New Group

Search conversations...

My status

Tap to add a status

Housekeeping Team

07:48 AM

You: Good morning team! Reminder: the daily...

Hotel 3

Adam Liwenga

07:48 AM

You: What should we order for the team lunc...

Hotel

Housekeeping Team

07:45 AM

You: Good morning team! Reminder: the daily...

Hotel 3

Housekeeping Team

07:41 AM

You: Good morning team! Reminder: the daily...

Hotel 3

Housekeeping Team

07:40 AM

You: Good morning team! Reminder: the daily...

Hotel 3

Housekeeping Team

07:40 AM

You: Good morning team! Reminder: the daily...

Hotel 3

AL

Adam Liwenga

Hotel MRK Grand Hotel

Good morning team! Reminder: the daily briefing is at 8:00 AM in the conference room.

You · 07:33 AM

Good morning team! Reminder: the daily briefing is at 8:00 AM in the conference room.

You · 07:35 AM

test plain

You · 07:35 AM

reverb test ok

You · 07:40 AM

Good morning team! Reminder: the daily briefing is at 8:00 AM in the conference room.

You · 07:40 AM

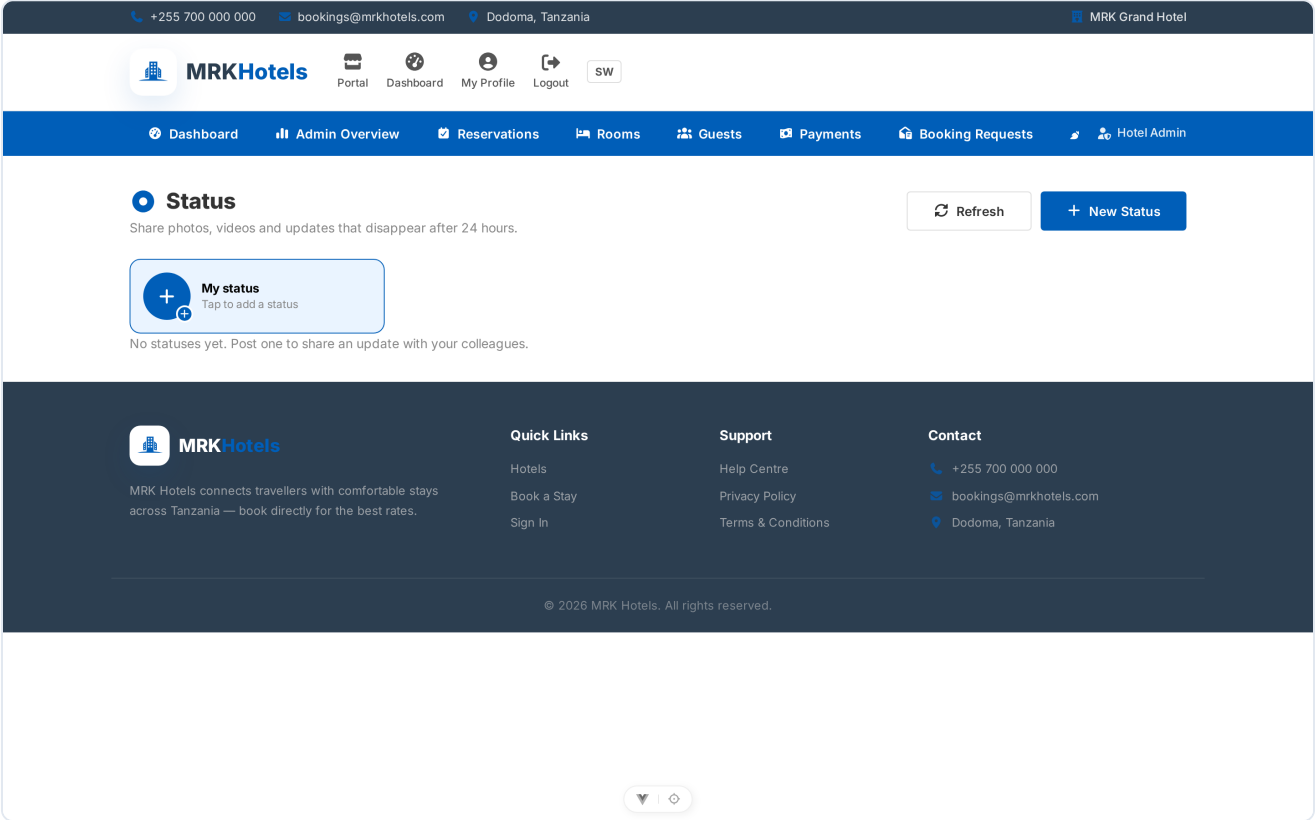
Good morning team! Reminder: the daily briefing is at 8:00 AM in the conference room.

I'll be there — saving you a seat next to the window.

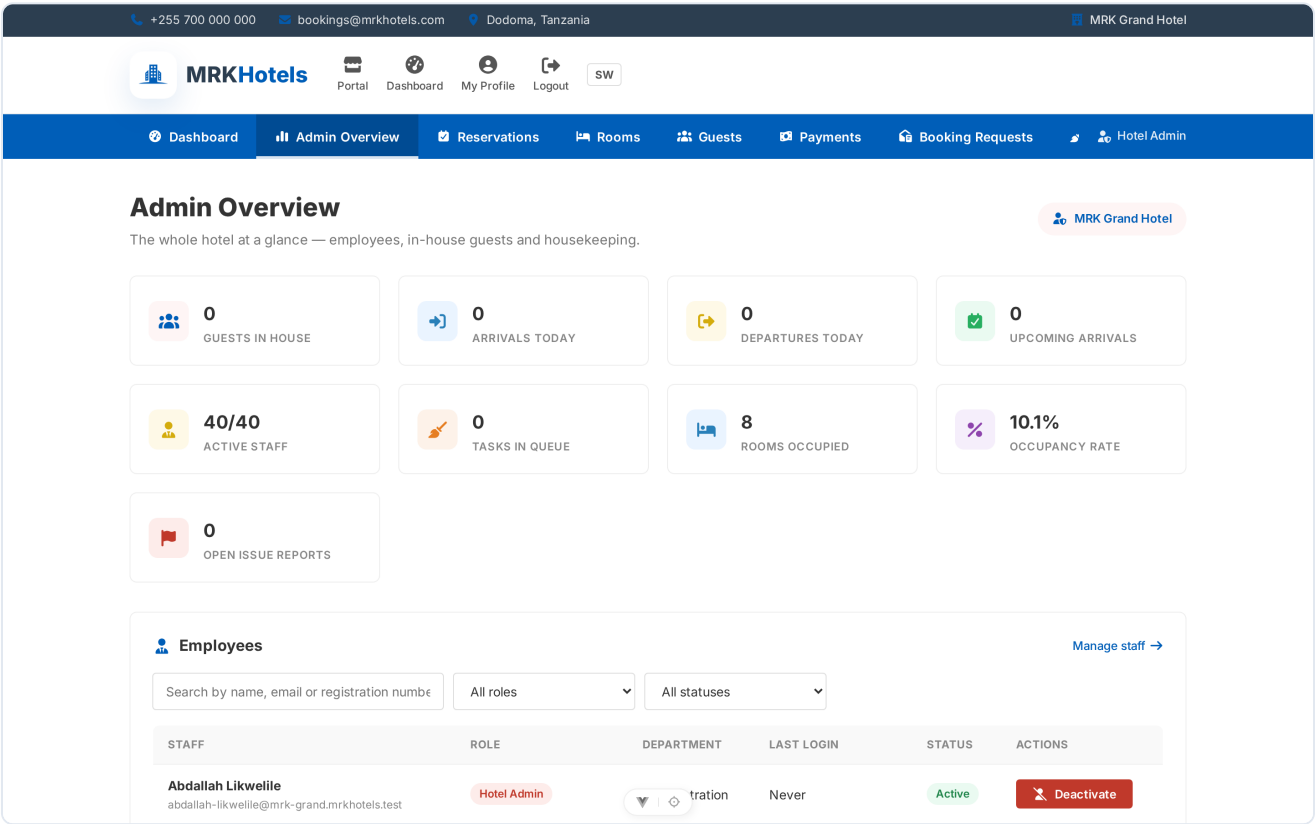
You · 07:40 AM

Messages.

Page 38 of 43



Statuses.



Admin overview.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

Reports

Occupancy, revenue and room status analytics.

Refresh

OverviewAudit logs

FromTo

08/04/202608/10/2026This weekThis month

OCCUPANCY (AVG)0%

REVENUE (PERIOD)TZS 420,000

TOTAL ROOMS79

REVENUE DAYS1

Room Status

TotalAvailableOccupiedCleaning

Revenue by method

CashTZS 420,000

Reports.

+255 700 000 000bookings@mrkhotels.comDodoma, TanzaniaMRK Grand Hotel

MRKHotels

PortalDashboardMy ProfileLogoutSW

DashboardAdmin OverviewReservationsRoomsGuestsPaymentsBooking RequestsHotel Admin

My Profile

Account

Hotel Admin

Hotel_admin

Active

Personal Information

First NameLast NameEmailPhoneID TypeID Number

HotelAdminadmin@mrkhotels.testTanzania (+255)None

PositionProfile Photo

Choose FileNo file chosen

Save

System Information

These details are managed by the hotel and cannot be edited.

Staff / Registration No.RoleRole LevelDepartmentHotelSub-manager

EMP-2026-0001hotelAdmin90administrationMRK Grand HotelNo

Last LoginMember Since

Profile.

Page 40 of 43

10.3 Superadmin panel

MRK Admin

Dashboard

Hotel Management

Reports

My Profile

Dashboard

Superadmin Victor Zacharia

Platform Overview

Monitor MRK Hotels platform activity.

22

TOTAL HOTELS

22

ACTIVE

0

PENDING APPROVALS

0

SUSPENDED

TZS 420,000

PLATFORM GMV (30 DAYS)

0

TRIALS EXPIRING (7D)

SaaS business

TZS 2,520,000

MRR (MONTHLY RECURRING)

TZS 30,240,000

ARR (ANNUAL RECURRING)

21

PAYING HOTELS

21

NEW SIGNUPS (30 DAYS)

0

ACTIVE TRIALS

100%

TRIAL → PAID CONVERSION

Platform GMV — last 6 months

Guest payments processed through the platform — not subscri...

Trials expiring soon

Platform Overview (`/superadmin`) — consolidated stats across all hotels.

MRK Admin

Dashboard

Hotel Management

Reports

My Profile

Hotel Management

Superadmin Victor Zacharia

Hotel Management

Onboard and manage hotels on the platform.

Refresh

New Hotel

Search

Status

Plan

Search hotels...

All

All

Clear

HOTEL	CONTACT	LOCATION	STATUS	PLAN	STAFF	ROOMS	ACTIONS
MRK Hotel Musoma mrk-musoma	MRK Hotels Group info@mrk-musoma.mrkhotels.test	Musoma, Tanzania	active	premium	31	73	Suspend
MRK Hotel Songea mrk-songea	MRK Hotels Group info@mrk-songea.mrkhotels.test	Songea, Tanzania	active	premium	31	73	Suspend
MRK Beach Resort Bagamoyo mrk-bagamoyo	MRK Hotels Group info@mrk-bagamoyo.mrkhotels.test	Bagamoyo, Tanzania	active	premium	31	73	Suspend
MRK Lodge Ngorongoro mrk-karatu	MRK Hotels Group info@mrk-karatu.mrkhotels.test	Karatu, Tanzania	active	premium	31	73	Suspend
MRK Hotel Mtwara mrk-mtwara	MRK Hotels Group info@mrk-mtwara.mrkhotels.test	Mtwara, Tanzania	active	premium	31	73	Suspend
MRK Hotel Singida mrk-singida	MRK Hotels Group info@mrk-singida.mrkhotels.test	Singida, Tanzania	active	premium	31	73	Suspend
MRK Hotel Shinyanga mrk-shinyanga	MRK Hotels Group info@mrk-shinyanga.mrkhotels.test	Shinyanga, Tanzania	active	premium	31	73	Suspend

Hotel Management — the tenant (hotel) directory.

Page 41 of 43

MRK Admin

Dashboard

Hotel Management

Reports

My Profile

Hotel Details

Superadmin Victor Zacharia

[← Back to Hotels](#)

MRK Beach Resort Bagamoyo

active

mrk-bagamoyo · Bagamoyo, Tanzania
info@mrk-bagamoyo.mrkhotels.test · 0755123456

STAFF
31

ROOMS
73

RESERVATIONS
0

OCCUPANCY
13.7%

REVENUE (30D)
TZS 0

Subscription

Plan: premium · Subscription Status: active · Trial ends: 8/8/2027

Plan

Subscription Status

Premium

Active

Update Subscription

Tax & invoice details

Printed on invoices issued to guests.

TIN

VRN

e.g. 123-456-789

e.g. 40-012345-6

Save tax details

Signature & stamp

PNG or JPG images printed on invoices above the authorize line.

Tenant detail — payment methods, tax IDs, branding.

MRK Admin

Dashboard

Hotel Management

Reports

My Profile

Reports

Superadmin Victor Zacharia

Reports


Reports is under construction.
This module will be built to match the Reports API endpoints.

Global reports across hotels.

MRK Admin

Dashboard

Hotel Management

Reports

My Profile

My Profile

Superadmin Victor Zacharia

My Profile

Account

Victor Zacharia

Superadmin

Active

Personal Information

First Name

Victor

Last Name

Zacharia

Email

superadmin@mrkhotels.test

Phone

Tanzania (+255)

ID Type

nationalId

ID Number

200011111411190000424

Position

Super Admin

Profile Photo

Choose File

No file chosen

Save

System Information

These details are managed by the hotel and cannot be edited.

Staff / Registration No.

EMP-2026-0011

Role

Superadmin

Role Level

100

Department

administration

Hotel

MRK Hotels Platform

Sub-manager

No

Last Login

2026-08-10 06:25

Member Since

2026-08-06

Superadmin profile.

End of developer documentation — © MRK Hotels.

Page 43 of 43